

Stopping Criteria Below the Signal-to-Noise Floor: Window Length, Not Tolerance, Governs Convergence Detection in Architecture Comparisons

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Architecture comparisons in efficient sequence modeling commonly stop training with a rule of the form “improvement in validation accuracy over the last w steps is below a tolerance τ ,” and then grant every architecture the resulting budget. We analyze one such rule that was pre-registered and frozen with a DOI before data collection, and show that it could not detect convergence—not because its tolerance was miscalibrated, but because the stopping decision was taken below the signal-to-noise floor of the measurement. In a multi-query associative recall benchmark, the delta-rule linear-attention condition improved at a rate of 0.31 accuracy points per evaluation in the regime treated as a plateau (positive in seven of eight seeds; $t = 4.20$, $p = .004$), while the noise of the tested difference was 1.16 points, giving a signal-to-noise ratio of 0.27. Under this ratio no tolerance separates the two errors of interest: rules that detect a plateau also stop during genuine improvement. We show that a proposed remedy—window means with a noise-calibrated tolerance—has a detection rate that is algebraically fixed at $\Phi(2)$ regardless of window length, and would have declared convergence while the model was still improving with probability .73. Decomposing the achievable gain, a factor of 7.0 comes from lengthening the comparison window and only 1.31 from replacing the two-point difference with a trend test. The failure is architecture-dependent: saturating conditions exhibit noise three orders of magnitude smaller and close immediately, so the shared budget covaries with the quantity under comparison. We give a power-based rule for choosing w that can be pre-registered without pilot data. All analyses reuse stored validation histories and require no additional training.

Keywords: stopping criterion, early stopping, statistical power, pre-registration, linear attention, associative recall, training budget, reproducibility

Introduction

Architecture comparisons in efficient sequence modeling rest on a decision that is rarely examined: when to stop training. A common practice is to train a reference architecture until a convergence rule fires and then grant every competing architecture the same number of steps (Li et al., 2019). The procedure is sound only if convergence can be detected reliably. If detection is unreliable, the shared budget inherits that unreliability, and so does the comparison.

This article reports a case in which detection was

impossible by construction, and characterizes why. The setting is a pre-registered comparison of attention head compositions on multi-query associative recall (MQAR; Arora et al., 2023). The stopping rule was specified, frozen, and published with a DOI before any model was trained. That matters for the argument developed here: the defect cannot be attributed to a rule chosen after seeing the results, because the rule could not have been chosen after seeing them.

Our diagnosis differs from the one an inspection of the rule alone would suggest. The tolerance was indeed smaller than the noise of the metric, and it is tempting to

conclude that the rule therefore measured noise instead of convergence. That conclusion does not survive contact with the data. The regime the rule was applied to was not a plateau: the architecture of interest was still improving, at a rate that a longer window would have detected. The rule failed because the *stopping decision* was taken below the signal-to-noise floor of the measurement, and no choice of tolerance repairs a decision taken below that floor.

Our contribution is fourfold. First, we show that the regime treated as converged carried a statistically detectable improvement trend, so the diagnosis must be framed in terms of signal-to-noise rather than tolerance. Second, we show that recalibrating the tolerance to measured noise—the natural remedy—produces a detection rate that is algebraically fixed and independent of the window, and buys detection power at the cost of stopping during genuine improvement. Third, we decompose the achievable gain and find that window length dominates the choice of test statistic by a factor of five. Fourth, we show that the noise driving all of this is architecture-dependent, which converts an apparently neutral procedural choice into a confound aligned with the variable under study, and we give a power-based rule for w that is pre-registrable without pilot data.

The Pre-Registered Setting

The experiment compares three head compositions in a small transformer (193,493 parameters) on MQAR: `softmax` (standard attention), `delta` (delta-rule linear attention), and `mix22` (two softmax heads and two delta

heads in the same layer). Delta-rule linear attention updates a fixed-size associative state by correcting the current read before writing (Yang et al., 2024); hybrid compositions are known to recover recall capacity that purely linear variants do not reach (Arora et al., 2023, 2024).

Validation accuracy was evaluated every 500 steps as the mean of top-1 accuracy over loads {64, 96, 128}, on a validation set held fixed across evaluations by a fixed random seed. The pre-registered rule, denoted D-004, declares convergence at step N when

$$\text{acc}(N) - \text{acc}(N - w) < \tau, \quad (1)$$

with $w = 500$ and $\tau = 0.005$. Because evaluations are spaced 500 steps apart, Equation 1 compares two *consecutive* evaluations. A condition closes its first training phase only when all eight seeds satisfy Equation 1; otherwise training proceeds in blocks of 2,500 steps up to a cap of 10,000. The common budget for the primary comparison is then the maximum of the per-condition convergence points, so one condition’s stopping behavior sets the budget granted to all.

Method

All analyses reuse validation histories stored during the original campaign; no model was retrained. The data comprise eight `delta` runs, eight `mix22` runs, and three `softmax` runs, plus one `delta` seed extended to 10,000 steps and used only where noted.

Noise and Trend Estimation

For each run we isolate the late-training regime—from step 4,000 for `delta` and 1,000 for the saturating conditions—and fit an ordinary least-squares line to validation accuracy against evaluation index. We estimate noise by two methods: the standard deviation of residuals from that fit, and the standard deviation of successive differences divided by $\sqrt{2}$. The second is unbiased under a linear trend and therefore serves as a check on the first. The quantity that governs Equation 1 is the standard deviation of the tested difference, $\sigma_D = \sigma\sqrt{2}$.

The trend is tested per run by the OLS slope b against $H_0: b = 0$, one-tailed, and across runs by a one-sample t test on the eight slopes.

 The author is an independent researcher, a student of the University Technical Degree in Programming (*Tecnica Universitaria en Programación*, TUP) at Universidad Tecnológica Nacional (UTN), Argentina, and a graduate of its Diploma in Artificial Intelligence. Correspondence concerning this article should be addressed to Maximiliano Speranza. E-mail: maximiliano.speranza@gmail.com. The stopping criterion analyzed here was pre-registered and frozen with a DOI *before* any data were collected, which rules out post hoc selection of the criterion as an explanation for the defect reported. The author declares no competing interests and received no funding for this work. Code, stored validation histories, and the analysis script that reproduces every number reported here are publicly available (see Open Practices).

Detection Probability

We report two estimates. The parametric estimate models late-training observations as $\text{acc}(N) = \mu + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, giving $P = \Phi(\tau/\sigma_D)$ for one seed and P^k for k independent seeds. The non-parametric estimate exploits the fact that Equation 1 with w equal to the evaluation interval is a successive difference: we therefore report the observed fraction of successive differences below τ , which requires no distributional assumption and no assumption of independence.

Results

Metric Noise Is Architecture-Dependent

Table 1 reports late-training noise by condition. The two estimators agree to within 1%, which also indicates that residual autocorrelation is negligible (see Assumption Checks). The decisive quantity is τ/σ_D : below 1, the tolerance cannot resolve the noise of the quantity it is compared against.

Table 1

Late-Training Noise of Validation Accuracy by Condition

Cond.	n	σ_{resid}	σ_{diff}	σ_D	τ/σ_D
delta	8	0.00821	0.00829	0.01162	0.43
mix22	8	0.00004	0.00004	0.00006	86.9
softmax	3	0.00004	0.00004	0.00006	88.8

Note. $\sigma_D = \sigma_{\text{resid}} \sqrt{2}$ is the standard deviation of the difference tested by Equation 1. Estimates use eight late-training evaluations per run for `delta` and four for the saturating conditions; the ratios for the latter should be read as “two orders of magnitude or more,” not as precise values.

The three conditions differ in σ by roughly three orders of magnitude. A single fixed tolerance is thus highly informative for two of them and unusable for the third.

The Regime Is Not a Plateau

Before asking whether the rule detects convergence, we ask whether there was convergence to detect. Table 2 tests the trend in the regime the rule was applied to.

The slope is positive in seven of eight seeds, with mean +0.00312 per evaluation; a one-sample t test on

Table 2

Trend Test in the Regime Treated as Converged (Steps 4,000–7,500)

Seed	b	t	p	Gain	D-004
0	+0.00259	1.86	.056	+1.81	yes
1	+0.00234	1.72	.068	+1.64	no
2	+0.00421	2.54	.022	+2.95	yes
3	+0.00334	5.12	.001	+2.34	no
4	−0.00014	−0.12	.547	−0.10	yes
5	+0.00445	3.58	.006	+3.11	no
6	+0.00141	1.37	.110	+0.98	yes
7	+0.00680	4.13	.003	+4.76	no

Note. b is the OLS slope per evaluation; p values are one-tailed against $H_0: b = 0$. “Gain” is the fitted improvement in accuracy points over the 3,500 steps of the window. The last column reports whether Equation 1 declared convergence at step 7,500.

the eight slopes rejects zero, $t(7) = 4.20$, $p = .004$. Extrapolating the mean slope predicts +1.56 accuracy points from step 7,500 to 10,000. The one seed actually extended to 10,000 gained +1.39 points of capacity at load 96 (0.9167 $\rightarrow$ 0.9306), consistent with the projection.

This reframes the problem. The `delta` condition had not converged at 7,500 steps. Driving it to the step cap was not an artifact of a noisy rule; it was the correct outcome reached for the wrong reason. The rule cannot be faulted for failing to detect a convergence that had not occurred—but neither can it be credited, since Table 2 shows its classifications bear no reliable relation to the measured trend.

The Decision Is Taken Below the Signal-to-Noise Floor

The two quantities can now be compared directly. The signal available to Equation 1 is one evaluation interval of improvement, $b = 0.00312$; the noise of the tested difference is $\sigma_D = 0.01162$. Their ratio is

$$\text{SNR} = \frac{b}{\sigma_D} = 0.27. \quad (2)$$

This single number governs everything that follows. A stopping rule must balance two errors: continuing to train after convergence, and stopping while improvement remains. With $\text{SNR} < 1$ the two error rates can-

not be separated by any choice of τ , because the distributions being discriminated overlap almost completely. Tolerance selection is not the lever; it merely slides a decision boundary along an axis on which the two hypotheses are not distinguishable.

Detection Probability

For `delta`, the parametric model gives $P = \Phi(0.43) = .667$ per seed and $P^8 = .039$ for all eight simultaneously. The non-parametric estimate—the observed fraction of successive differences below τ in the same regime, $N = 56$ —gives $P = .554$, 95% CI [.423, .684], and $P^8 = .009$. The non-parametric estimate is preferable: it assumes nothing, and it fits the observation better. Four of eight seeds were classified as converged; the expected count is 4.4 under the non-parametric estimate ($P(X \leq 4) = .52$) versus 5.3 under the parametric one ($P(X \leq 4) = .26$).

Either way, the condition had less than a 4% chance of ever closing its phase by convergence—which is the correct outcome here, since it had not converged, but would have been equally true had it converged.

Recalibrating the Tolerance Does Not Help

The natural remedy is to compare window means rather than points, and to set the tolerance from measured noise:

$$\overline{\text{acc}}_{\text{last } m} - \overline{\text{acc}}_{\text{prev. } m} < c \sigma_D^{(m)}, \quad \sigma_D^{(m)} = \sigma \sqrt{2/m}. \quad (3)$$

With $m = 3$ and $c = 2$, the all-seed detection probability rises from .039 to .832. That figure is not evidence of a better rule. Setting the tolerance to $c \sigma_D^{(m)}$ makes the per-seed detection rate equal to $\Phi(c)$ *identically*, for any m and any σ : with $c = 2$ it is $\Phi(2) = .977$ whatever the window. Table 3 separates the two ingredients on our data. The entire gain comes from relaxing the tolerance, not from windowing.

The cost of that relaxation is the error the detection rate does not measure. Under a true improvement rate b , the probability that Equation 3 declares convergence anyway is $\Phi(c - b m / \sigma_D^{(m)})$. At the improvement rate actually measured, that probability is .73: the proposed rule would have stopped `delta` while it was still gaining roughly one accuracy point per window. Its minimum detectable improvement, at a 5% false-stop rate, is

4.07 points per 2,500-step block—larger than the 1.39-point gain that the extended seed actually realized over such a block. A rule tuned to detect plateaus reliably in this regime is, necessarily, a rule that stops during real progress.

Window Length Dominates the Test Statistic

What does raise the floor is the window. For n equally spaced evaluations, the OLS slope has standard error

$$\text{SE}(b) = \sigma \sqrt{\frac{12}{n(n^2 - 1)}}, \quad (4)$$

which falls as $n^{-3/2}$, against $n^{-1/2}$ for averaging. On our data, with $n = 8$ evaluations, $\text{SE}(b) = 0.00127$ and the measured mean slope yields $t = 2.46$: the same eight points that leave Equation 1 at $\text{SNR} = 0.27$ support detection when used jointly. The total gain of $9.2\times$ decomposes into $7.0\times$ from comparing points seven evaluations apart instead of one, and $1.31\times$ from using the trend statistic instead of the endpoint difference. Window length, not the choice of statistic and not the tolerance, is the dominant term.

This yields a rule that can be fixed in advance. Given a smallest improvement rate b^* that the study commits to not overlooking, a one-tailed level α , and power $1 - \beta$, the window must satisfy

$$b^* \geq (z_\alpha + z_\beta) \sigma \sqrt{\frac{12}{n(n^2 - 1)}}. \quad (5)$$

Table 4 evaluates Equation 5 at the noise level measured here. It shows what the original design would have required: detecting an improvement of one accuracy point per 2,500 steps needs about eleven evaluations, whereas D-004 used two.

Equation 5 needs σ , which is not known before training. Two routes avoid circularity. Where the validation set is resampled at each evaluation, sampling error gives an analytic lower bound, $\sigma \geq \sqrt{p(1-p)/n_{\text{val}}}$, computable at design time. Where it is held fixed, as here, that term vanishes and the residual noise is optimizer variability, which must be measured; a short pilot on any one condition suffices, and—unlike retuning a tolerance after seeing results—it fixes a window, not a verdict.

Table 3*Decomposition of the Detection Gain*

Variant	τ	σ_D	P^8
D-004 as pre-registered	0.0050	0.01162	.039
Windowing only ($m = 3$, τ fixed)	0.0050	0.00671	.126
Retuned tolerance only ($2\sigma_D$)	0.0232	0.01162	.832
Equation 3 in full	0.0134	0.00671	.832

Note. P^8 is the probability that all eight seeds are simultaneously declared converged while on a plateau. The two lower rows coincide because setting $\tau = 2\sigma_D$ fixes the per-seed rate at $\Phi(2)$ irrespective of the window.

Table 4*Detectable Improvement as a Function of Window Length*

n	Window	SE(b)	b^*
4	1,500	0.00367	4.56
6	2,500	0.00196	2.44
8	3,500	0.00127	1.58
10	4,500	0.00090	1.12
12	5,500	0.00069	0.85
16	7,500	0.00045	0.55
20	9,500	0.00032	0.40

Note. n = number of evaluations in the window; “Window” is its span in training steps. $\sigma = 0.00821$, $\alpha = .05$ one-tailed, power = .80. b^* is the smallest improvement rate detectable with that power, in accuracy points per 2,500 steps.

Propagation to the Training Budget

Because the rule sets the common budget, its architecture-dependence is not neutral. Saturating conditions have $\tau/\sigma_D \approx 87$ and close at the first opportunity; the non-saturating condition is still improving and is driven to the cap. Whichever budget is then adopted is either insufficient for the condition that is still learning or far beyond what the saturating ones need, and *which of the two occurs is decided by the conditions that saturate*. The granted budget thus covaries with task performance—the very quantity under comparison.

The magnitude is material. The pre-registered decision margin for the primary hypothesis was 2 accuracy points; the seed extended from 7,500 to 10,000 steps gained 1.39 points of capacity at load 96. A budget-induced difference of the same order as the decision margin is not a rounding concern. This estimate rests

on a single extended seed and is indicative rather than conclusive; the trend analysis of Table 2, which uses all eight, supports the same direction and magnitude.

Assumption Checks

Residual autocorrelation. Lag-1 autocorrelation of detrended residuals is negative in seven of eight runs, median -0.315 , which invites a correction to $\text{Var}(D)$. The correction would be spurious. With $n = 8$ points and a fitted trend removed, the lag-1 estimator is strongly biased downward even for white noise: simulation under white noise (20,000 replicates) gives $E[\hat{\rho}_1] = -0.287$, 95% interval $[-0.803, +0.310]$, placing the observed median at the 49th percentile of the null. This is the small-sample bias of autocorrelation estimators (Marriott & Pope, 1954). The data contain an independent check: were $\rho_1 = -0.315$ real, $\sigma_{\text{diff}}/\sigma_{\text{resid}}$ would equal $\sqrt{1 - \rho_1} = 1.147$; the observed ratio is 1.010, implying $\rho_1 = -0.02$. We therefore treat the residuals as serially uncorrelated. We note the point because the tempting correction runs in the direction that flatters the argument, and would have been reported as support for it.

Source of the noise. The validation set is held fixed across evaluations by a fixed seed, and comprises roughly 37,000 queries; resampling error would contribute $\sigma \approx 0.0018$ and, being fixed, contributes none. The measured $\sigma = 0.0082$ is therefore attributable to optimizer variability, not to measurement error. Enlarging the validation set would not reduce it; temporal averaging or weight averaging would.

Choice of regime onset. Varying the onset from step 2,500 to 5,000 leaves τ/σ_D between 0.39 and 0.51, always well below unity, and leaves the sign and significance of the trend unchanged.

Does validation accuracy track the quantity of interest? Across the eight δ seeds, final validation accuracy correlates with capacity at load 96 at $r = .96$ and at load 128 at $r = .99$. The metric is not blind to the axis under study; it is noisy relative to its own rate of change. The problem is one of power, not of construct.

Discussion

Implications

The practical cost of a stopping rule below the noise floor is not primarily wasted computation, although in our campaign roughly 80% of the projected budget was attributable to it. The serious cost is that the training budget becomes a function of the metric’s noise, which is itself a function of where the architecture sits on the task. Architectures that saturate stop early; architectures in the interesting intermediate regime are driven to the cap. Any comparison drawn across that asymmetry is confounded, and the confound is aligned with the variable the study exists to measure.

The episode also carries a lesson about pre-registration. Freezing a stopping rule is good practice, but a rule frozen without a power analysis freezes an unexamined assumption about the noise of the metric. Our pre-registration fixed τ and w and said nothing about σ ; Equation 5 is what should have been registered instead, since it states what the study commits to detecting rather than how it will compute a difference. This is the analogue, for training-time decisions, of registering a target effect size rather than a p -value threshold.

Relation to Prior Work

That validation-based early stopping is unstable in noisy regimes is established (Prechelt, 1998), as is the need to control training budget when comparing architectures (Li et al., 2019), and seed-sampling noise has been shown to affect the reliability of synthetic benchmarks (Chanin, 2026). The specific contribution here is the chain that links them and its quantification: a stopping decision taken below the signal-to-noise floor, a floor that differs by three orders of magnitude across the architectures being compared, and a budget rule that transmits that difference into the comparison. We also report, as a cautionary detail, that the two natural repairs—retuning the tolerance to measured noise,

and averaging over a window—address the smaller term while leaving the dominant one untouched.

Limitations

This is a single documented case, not a general study. It involves one task family (MQAR), one model scale (193K parameters), one hardware platform, three conditions, and eight seeds per condition. We do not claim that fixed-tolerance rules fail generally, nor that any published result is invalidated by this mechanism. The budget-effect estimate at load 96 rests on one extended seed; the trend analysis that corroborates it uses all eight but extrapolates beyond the observed range. The power rule of Equation 5 assumes an approximately linear improvement trend and homoscedastic noise over the window, which is an approximation for late-training dynamics and would fail near a sharp transition. Finally, Equation 5 is derived here and validated only retrospectively; prospective validation remains future work, and is the natural next step for this line.

Recommendations

We suggest four inexpensive practices. First, before fixing a stopping rule, state the smallest improvement rate the study commits to detecting, and size the window from it via Equation 5; report the resulting power. Second, report τ/σ_D rather than τ alone, since the tolerance is uninterpretable without the noise of the quantity it is compared against. Third, report noise per condition: a rule adequate for one architecture may be unusable for another in the same study, and that asymmetry is a confound, not an inconvenience. Fourth, when a rule is relaxed to raise its detection rate, report the complementary error—the probability of stopping under the smallest improvement that matters—because detection rate alone can always be driven to any desired value.

Conclusion

A stopping rule compares an improvement against a tolerance, but what determines whether it can work is neither of those quantities in isolation: it is the ratio between the improvement accumulated over the comparison window and the noise of the comparison. When that ratio is below unity, the rule reports which way the last evaluation happened to fall, and no tolerance recovers the decision. Lengthening the window does, at a rate

that makes the fix cheap. When such a rule also sets the shared training budget of an architecture comparison, its architecture-dependent noise sensitivity converts a procedural choice into a confound aligned with the variable under study. Diagnosing this required nothing beyond validation histories that were already being stored, and correcting it requires no additional training.

Open Practices

The pre-registration was frozen and published before data collection (DOI: 10.5281/zenodo.21495252). Code, stored validation histories, and the analysis script that reproduces the reported numbers are available at <https://github.com/SperanzaMax/telar-ligamento>; the analysis runs on CPU in under a minute. One exception must be stated: the single `delta` seed extended to 10,000 steps, cited once for its capacity gain of 1.39 points, was lost to a session failure before it could be versioned, and is the only quantity here that cannot be recomputed from the repository. Every other figure, including the trend analysis that independently predicts that gain, derives from the eight versioned runs.

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